

HASHEM ALOMAR

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Education

University of Maryland, College Park

May 2028

Combined Bachelor & Master of Science in Computer Science and Mathematics

GPA: 3.9

Relevant Coursework: Algorithms, Data Structures, Discrete Mathematics, Real Analysis, Statistics, Probability Theory, Linear Algebra, Multivariable Calculus, Object-Oriented Programming, Competitive Programming, Financial Mathematics

Experience

University of Maryland - Furong Lab

August 2026 – Present

Research Assistant

- Exploring LLM attention sinks across transformer layers as a mechanism to create a training-free optimization for loop transformers.
- Researching agentic interpretability tools as an intervention method for agent failures.

Stripe

May 2026 – August 2026

Software Engineering Intern

- Design an end-to-end real-time validation system for change event streams, enabling detection of **10,00,000+** missing or incorrect events per month in O(minutes).
- Architect a change event aggregator in Go that ingests write-ahead logs at **100k RPS** peak throughput.
- Design a battle-tested alert system and run book to help engineers debug missing data in under **20 minutes**.

University of Maryland

January 2025 – May 2026

Research Assistant - Quantum Machine Learning

- Optimized QLSTM circuitry using **PennyLane**, to predict 10+ equities concurrently, **saving ~\$100 per iteration**.
- Achieved **7% higher model accuracy** by testing QLSTM models on **IonQ** hardware across 3–8 qubits.
- **Reduced RMSE (error) by 12%** by optimizing QLSTM circuit depth and topology, leveraging IonQ API.

GeneticFit Labs

September 2024 – March 2025

Chief Technology Officer

- Founded and served as CTO of a genetic training startup that raised \$150K in seed funding, with a **\$3M valuation**.
- Built and deployed an iOS MVP in **Swift and Firebase**, tested by 20+ pilot users, **with a 95% satisfaction rate**.
- **Led a 10+ team** of software engineers and designers from ideation through full-stack deployment.

Projects & Extracurriculars

Apex Fund at UMD — Quantitative Analyst

January 2025 – Present

- Led the design and deployment of systematic trading strategies using machine learning and quantitative models, improving risk adjusted return by 10% for \$10,000 AUM portfolio.
- Building a mid-frequency crypto trading model across 7+ exchanges in a 5-person team, generating \$200 weekly.
- Built an automated trading bot that exploits real-time odds mispricing between Kalshi and major sportsbooks, executing positions via Kalshi's API before market convergence, generating \$500 daily.

ProjectLIFT — Founder

April 2025 – Present

- Founded a student-run club delivering tech solutions to 10+ local small businesses, saving over \$200,000+ in expenses.

Math Club — Officer

August 2024 – Present

- Leading weekly problem-solving sessions and discussions across diverse mathematical topics in club with 100+ students.
- Tutoring 100+ students in Calculus, Linear Algebra, Statistics and Differential Equations.

Skills, Awards & Interest

Programs: Assembly, Python, Java, C, C++, SQL, Pytorch, NumPy, Pandas, TensorFlow, Scikit Learn, PennyLane, Go

Awards: Bloomberg BPuzzled 2nd Place, USA Climbing Divisionals: 2nd Place, Jordanian Karate National Champion(Youth)

Interests: Poker, Rock Climbing, Hiking, Puzzles, Blackjack, Chicago Bulls, Philosophy